

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I K.Sujana Sree(192525320) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.Sujana Sree

Annexure A

Comparative Analysis

1. Apply **SSL/TLS and IPsec protocols** to a secure communication scenario and determine which provides better security for web-based and network-layer protection.

ANSWER:

1. Introduction

SSL/TLS and **IPsec** are both security protocols, but they protect communication at different network layers.

- **SSL/TLS** operates mainly at the **transport/application interface** and is commonly used to secure web traffic through **HTTPS**.
 - **IPsec** operates at the **network layer (Layer 3)** and can secure IP packets between hosts, routers, or entire networks.
-

2. Secure Communication Scenario

Consider a company with:

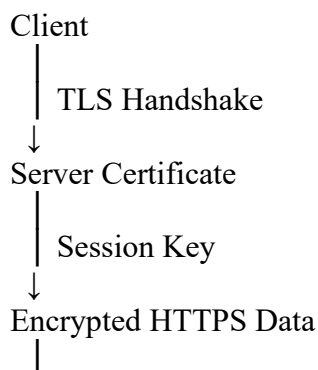
- Employees accessing an online banking/web application.
- Two company branches communicating through the Internet.
- An attacker monitoring Internet traffic.

Scenario 1: Web Application

An employee accesses:

`https://company.com`

TLS establishes a secure connection:



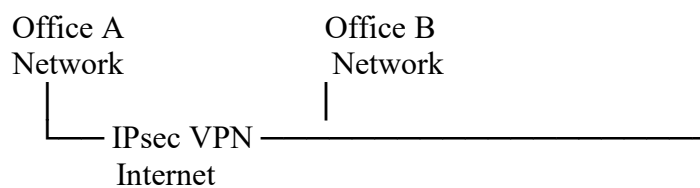
↓
Web Server

TLS provides:

- Server authentication
- Confidentiality
- Data integrity
- Protection against eavesdropping

Scenario 2: Branch-to-Branch Network

Two offices communicate over the Internet:



IPsec creates a secure tunnel between the networks.

It can protect:

- IP addresses within the protected tunnel
- Application data
- TCP/UDP packets
- Different types of IP traffic

3. SSL/TLS vs IPsec

Feature	SSL/TLS	IPsec
Primary layer	Transport/Application interface	Network Layer
Main use	HTTPS, secure web applications	VPNs and IP communication
Protection	Application traffic	IP packets
Authentication	Certificates	Certificates/PSK/IKE authentication
Encryption	Yes	Yes
Integrity	Yes	Yes
Web security	Excellent	Good, but not application-specific
Network-wide protection	Limited	Excellent
VPN support	Often application-specific	Strong
Deployment	Usually easier for web services	More complex

Feature	SSL/TLS	IPsec
Typical example	HTTPS	Site-to-site VPN

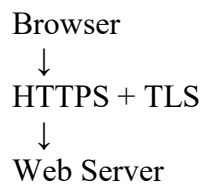
4. Which Provides Better Security?

There is **no universal winner** because they protect different layers.

For Web-Based Protection → SSL/TLS

TLS is better for web applications because it directly secures the communication between the browser and web server.

For example:



It provides:

- Server authentication through certificates
- Confidentiality
- Integrity
- Protection of application data

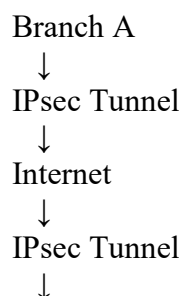
Therefore:

SSL/TLS is better for web-based protection

For Network-Layer Protection → IPsec

IPsec is better when an organization needs to secure **IP traffic across an entire network**.

For example:



Branch B

Applications do not necessarily need to be modified because IPsec protects traffic at the network layer.

Therefore:

IPsec is better for network-layer protection

5. Security Comparison

SSL/TLS

Advantages:

- Excellent for HTTPS
- Strong server authentication
- Protects application data
- Easy integration with modern web browsers
- Supports modern protocols such as TLS 1.3

Limitation:

- It normally protects specific application connections rather than automatically protecting every IP packet generated by a host or network.

IPsec

Advantages:

- Protects IP packets at the network layer
- Can secure entire networks through VPN tunnels
- Application-independent
- Suitable for site-to-site communication

Limitation:

- More complex to configure and manage.
 - Does not replace application-layer authentication or security controls.
-

6. Final Recommendation

Requirement

Recommended Protocol

Requirement	Recommended Protocol
Secure website/HTTPS	SSL/TLS
Browser-to-server communication	SSL/TLS
Secure email/web application connection	SSL/TLS
Site-to-site VPN	IPsec
Branch-to-branch communication	IPsec
Protect entire IP network	IPsec
Network-layer security	IPsec

Conclusion

SSL/TLS is the better choice for web-based protection because it securely authenticates and encrypts application communication between clients and servers. IPsec is better for network-layer protection because it can secure complete IP traffic between hosts or entire networks.

In an enterprise environment, **both can be used together**: IPsec can secure the network connection between branches, while TLS can provide additional end-to-end security for web applications.

2. Design a secure email system by comparing **end-to-end encryption and gateway-based encryption**, and justify the most suitable approach for enterprise use.

ANSWER:

A secure email system can use either **end-to-end encryption (E2EE)** or **gateway-based encryption**.

1. End-to-End Encryption

In E2EE, the email is encrypted by the sender and decrypted only by the intended receiver.

Advantages:

- Strong confidentiality.
- Even email servers cannot read the message.
- Protects sensitive business information.

Disadvantages:

- Key management is difficult.
- Searching, filtering, and malware scanning become harder.
- Users must manage encryption keys.

2. Gateway-Based Encryption

In gateway-based encryption, an organization's email gateway encrypts and decrypts messages before sending or receiving them.

Advantages:

- Centralized security management.
- Easier key management.
- Supports antivirus, spam filtering, and content inspection.
- Suitable for large organizations.

Disadvantages:

- Gateway can access plaintext messages.
- Compromise of the gateway can expose emails.
- Provides less end-to-end confidentiality.

Comparison

Feature	End-to-End Gateway-Based	
Confidentiality	Very High	High
Key Management	Difficult	Easier
Centralized Control	Low	High
Content Inspection	Difficult	Easy
Enterprise Management	Complex	Easy

Recommendation

For **enterprise use**, **gateway-based encryption is generally more practical** because it provides centralized control, easier key management, and integration with security systems. However, for highly sensitive information such as financial, legal, or confidential communications, **end-to-end encryption is preferable**.

Conclusion: A hybrid approach is ideal—use **gateway-based encryption for general enterprise email** and **end-to-end encryption for highly confidential messages**.

3. Apply **HMAC and Digital Signature techniques** to a data transmission scenario and analyze which method provides better authentication and integrity.

ANSWER:

HMAC (Hash-based Message Authentication Code) and **Digital Signatures** provide **authentication and data integrity**, but they use different types of keys.

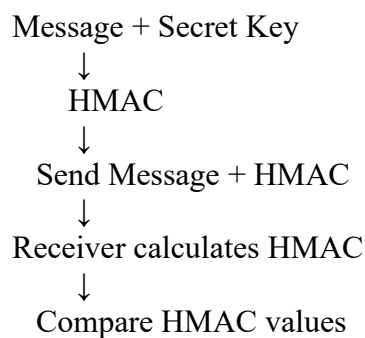
Scenario

Consider a company sending confidential data from **Sender A** to **Receiver B**.

1. Using HMAC

- Sender and receiver share the same **secret key**.
- Sender generates an HMAC using the message and secret key.
- Receiver uses the same key to calculate the HMAC.
- If both HMAC values match, the message is authentic and unchanged.

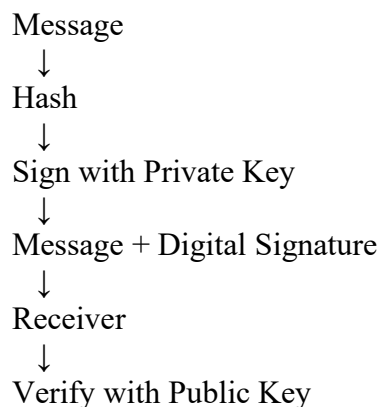
Process:



2. Using Digital Signature

- Sender generates a **private/public key pair**.
- Sender signs the message using the private key.
- Receiver verifies the signature using the sender's public key.
- A valid signature confirms the message's integrity and sender authentication.

Process:



Comparison

Feature	HMAC	Digital Signature
Key type	Shared secret key	Private/Public key pair

Feature	HMAC	Digital Signature
Integrity	Yes	Yes
Authentication	Yes	Yes
Non-repudiation	No	Yes
Speed	Faster	Slower
Key distribution	Difficult for many users	Easier using public keys
Best use	Internal trusted systems	Internet/enterprise communication

Which Is Better?

HMAC is faster and efficient when sender and receiver already share a secret key. However, managing secret keys becomes difficult in large systems.

Digital signatures provide stronger authentication because the sender's private key is unique and the public key can be distributed openly. They also provide **non-repudiation**, which HMAC cannot provide because both parties know the same secret key.

Conclusion

For **large-scale or Internet-based communication**, **Digital Signatures provide better authentication and non-repudiation**, while **HMAC provides faster integrity and authentication when a shared secret key is available**. Therefore, digital signatures are generally more suitable when strong identity verification is required.

4. Create a **spam detection system** by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.

ANSWER:

HMAC (Hash-based Message Authentication Code) and **Digital Signatures** provide **authentication and data integrity**, but they use different types of keys.

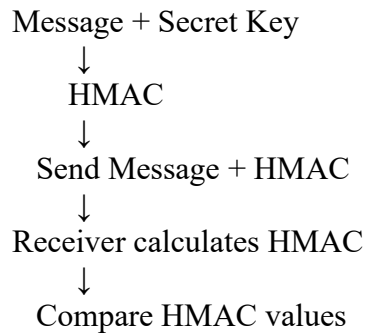
Scenario

Consider a company sending confidential data from **Sender A** to **Receiver B**.

1. Using HMAC

- Sender and receiver share the same **secret key**.
- Sender generates an HMAC using the message and secret key.
- Receiver uses the same key to calculate the HMAC.
- If both HMAC values match, the message is authentic and unchanged.

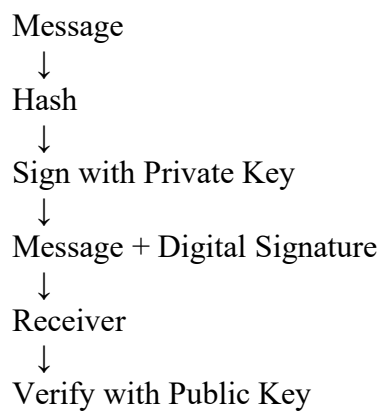
Process:



2. Using Digital Signature

- Sender generates a **private/public key pair**.
- Sender signs the message using the private key.
- Receiver verifies the signature using the sender's public key.
- A valid signature confirms the message's integrity and sender authentication.

Process:



Comparison

Feature	HMAC	Digital Signature
Key type	Shared secret key	Private/Public key pair
Integrity	Yes	Yes
Authentication	Yes	Yes
Non-repudiation	No	Yes
Speed	Faster	Slower
Key distribution	Difficult for many users	Easier using public keys
Best use	Internal trusted systems	Internet/enterprise communication

Which Is Better?

HMAC is faster and efficient when sender and receiver already share a secret key. However, managing secret keys becomes difficult in large systems.

Digital signatures provide stronger authentication because the sender's private key is unique and the public key can be distributed openly. They also provide **non-repudiation**, which HMAC cannot provide because both parties know the same secret key.

Conclusion

For **large-scale or Internet-based communication**, **Digital Signatures provide better authentication and non-repudiation**, while **HMAC provides faster integrity and authentication when a shared secret key is available**. Therefore, digital signatures are generally more suitable when strong identity verification is required.

5. Design a secure communication framework by comparing **AES and RSA encryption techniques**, and determine their roles in achieving confidentiality and efficiency.

ANSWER:

AES and RSA are commonly used together in secure communication because they have different roles. AES provides fast data encryption, while RSA provides secure key exchange and authentication.

1. AES Encryption

AES is a **symmetric encryption** technique, meaning the same secret key is used for encryption and decryption.

Advantages:

- Very fast.
- Suitable for encrypting large files and messages.
- Requires less computational power.

Limitation:

- Securely sharing the secret key between sender and receiver is difficult.

2. RSA Encryption

RSA is an **asymmetric encryption** technique that uses a public key and a private key.

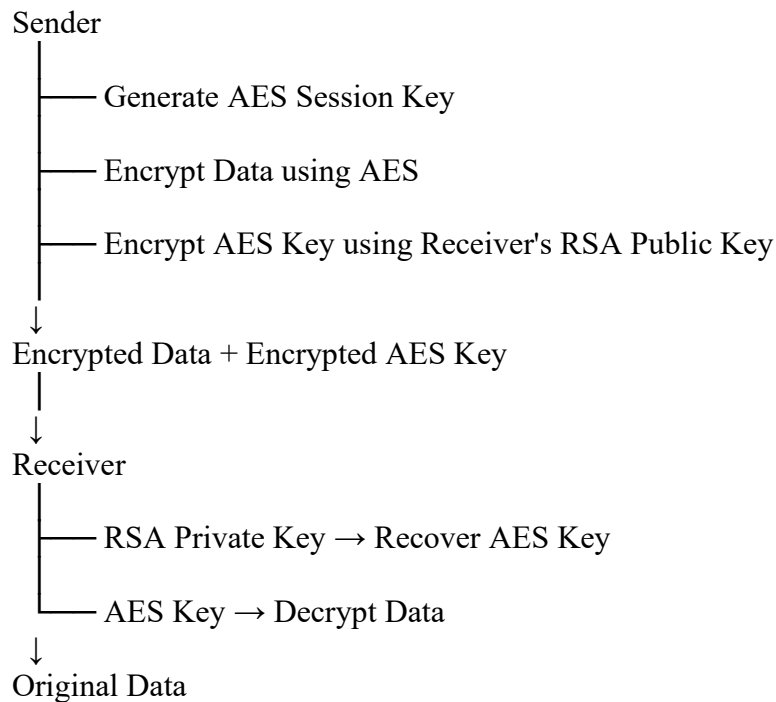
Advantages:

- Solves the key-distribution problem.
- Public key can be shared openly.
- Can be used for digital signatures and authentication.

Limitation:

- Slower than AES.
- Not efficient for encrypting large amounts of data directly.

Secure Communication Framework



Comparison

Feature	AES	RSA
Type	Symmetric	Asymmetric
Keys	One shared key	Public + private key
Speed	Very fast	Slower
Large data encryption	Suitable	Not suitable
Key exchange	Difficult	Suitable
Authentication	Limited	Supports digital signatures
Main role	Data encryption	Key exchange/authentication

Conclusion

A **hybrid approach** provides the best solution. **AES encrypts the actual data efficiently**, while **RSA securely exchanges the AES session key and can provide authentication through digital signatures**. Thus, AES provides **efficiency and confidentiality**, while RSA provides **secure key distribution and authentication**.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured	Generally clear with minor issues	Somewhat unclear or poorly	Disorganized and difficult to

		presentation		structured	understand
--	--	--------------	--	------------	------------